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Paleomagnetism of Eocene red-beds in the eastern part of the Qiangtang Terrane and its implications for uplift and southward crustal extrusion in the southeastern edge of the Tibetan Plateau

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ABSTRACT

A primary magnetic component with the acquisition time of 56.0–43.2 Ma was isolated between 580 °C and 685 °C from the Eocene Gonjo and Ranmugou Formations in the eastern part of the Qiangtang Terrane, Tibetan Plateau. The tilt corrected site-mean direction is $D_s = 35.5°$, $I_s = 29.3°$, $k = 45.9$, $\alpha_{95} = 3.2°$. The site-mean inclination increased from 29.3° to 41.6° after multiple inclination shallowing corrections, giving a paleopole of 57.9°N/192.1°E, $A_{95} = 2.9°$. Comparison of the Paleogene paleomagnetic data for the Qiangtang Terrane and Lhasa Terrane reveals that both terranes experienced latitudinal crustal shortening before 54–43 Ma, indicating the uplift of southern and central Tibet in the early Eocene. Subsequently, since 35.4 ± 2.4 Ma, the north of the Qiangtang Terrane experienced ~1300 ± 410 km of crustal shortening, indicating the uplift of the northern Tibetan Plateau. The Lhasa Terrane and Qiangtang Terrane have not experienced further crustal shortening since the late Eocene, and the southeastern part of Tibet cannot have provided abundant crustal material to accommodate the significant crustal southward extrusion in the southeastern edge of the Tibetan Plateau. The crust of the Tengchong Terrane and Shan–Thai Block did not experience significant southward extrusion since the late Oligocene–Early Miocene. The Indochina Block was situated in the north of the Qiangtang Terrane before the Oligocene, and since the early Oligocene, the Indochina Block began to experience southward extrusion from the north of the Qiangtang Terrane, which absorbed part of the crustal shortening in the north of the Qiangtang Terrane.

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1. Introduction

The collision of the Indian Plate and Eurasia in the Paleocene, and the subsequent extrusion between them induced significant latitudinal crustal shortening across southern Eurasia during the Cenozoic (Tapponnier et al., 1982; Peltzer and Tapponnier, 1988; Jaeger et al., 1989; Beck et al., 1995; Yin and Harrison, 2000; Cande and Stegman, 2011; Van Hinsbergen et al., 2011). Part of the latitudinal crustal shortening (~900 km) was absorbed by the Cenozoic fold and thrust belts system across the Tibetan Plateau (Yin and Harrison, 2000; Spurlin et al., 2005; Yin et al., 2007; Van Hinsbergen et al., 2011). Examples are the Shiquanhe–Gaize–Amdo Thrust system in the southern part of the Qiangtang Terrane (QT), the Fenghuoshan–Nangqian fold belt in the northeastern part of the QT, the Qimen Tagh–North Kunlun thrust system, and the Nan Shan thrust system in the Qaidam Basin (Meyer et al., 1998; Yin and Harrison, 2000; DeCelles et al., 2002; Spurlin et al., 2005; Yin et al., 2007; van Hinsbergen et al., 2011). The other part is believed to have been accommodated since the Oligocene by large-scale southward extrusion of the Tengchong Terrane (TCT), the Shan–Thai Block (STB) and the Indochina Block (ICB), in the southeastern edge of the Tibetan Plateau, along several lithospheric-scale strike-slip fault systems (Tapponnier et al., 1990; Yang and Besse, 1993; Yang et al., 1995; Leloup et al., 1995; Wang et al., 1998a, 1998b; Wang and Burchfiel, 2000; Molnar and Dayem, 2010).

Recent GPS observations in the interior of Tibet and the surrounding areas show that the crust of the southeastern part of

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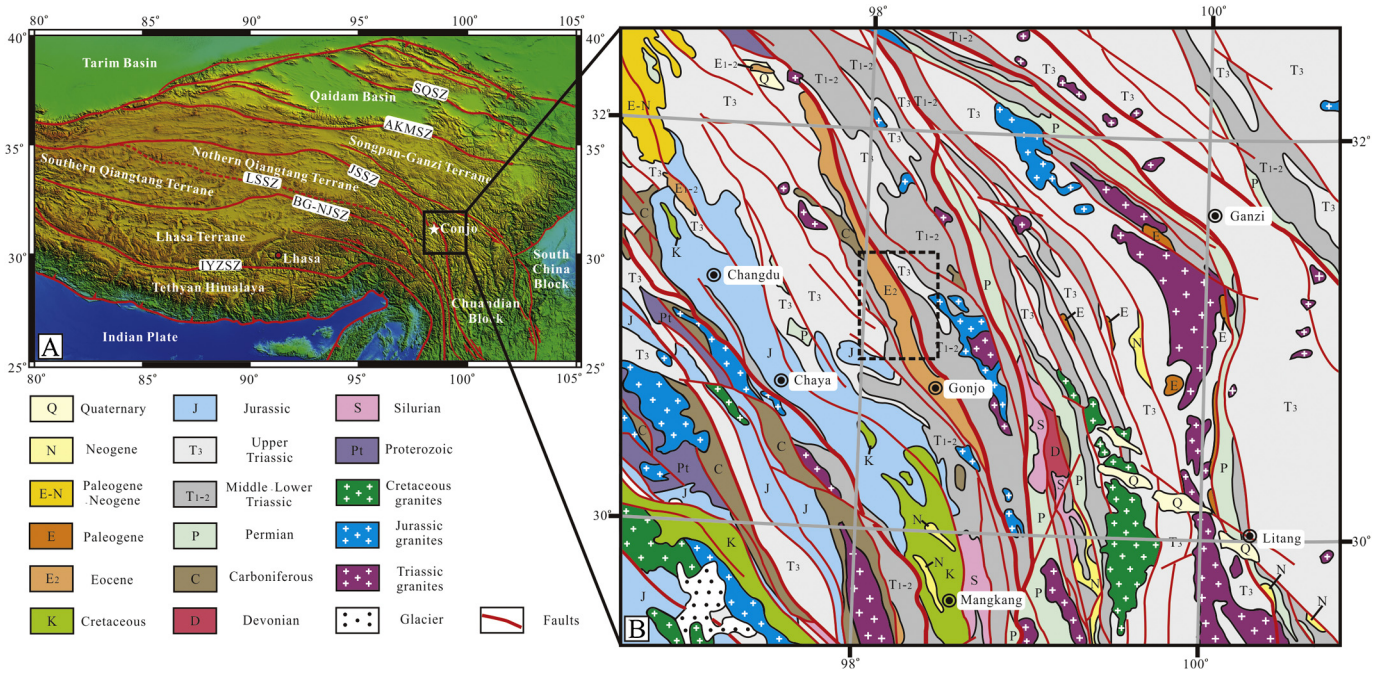


Fig. 1. (A): A schematic tectonic geological map of the Tibetan Plateau. (B): Tectonic map of the Gonjo area. IYZSZ, Indus-Yarlung Zangbo Suture Zone. BG-NJSZ, Bangong–Nujiang Suture Zone. LSSZ, Longmu Tso-Shuanghu Suture Zone. JSSZ, Jinshajiang Suture Zone. AKMSZ, Ayimaqing–Kunlun–Muztagh Suture Zone. SQSZ, South Qilian Suture Zone.

the Tibetan Plateau is currently experiencing clockwise rotational extrusion around the Eastern Himalayan Syntaxis (EHS) (Chen et al., 2000; Shen et al., 2005; Gan et al., 2007). Paleomagnetic studies of the Cretaceous and Paleogene red-beds of the ICB indicate that it experienced clockwise rotation accompanied by ~700 ± 200 km southward extrusion since the Oligocene, which accommodated part of the post-collision crustal shortening across southern Eurasia (Yang et al., 1995; Otofuji et al., 1998; Sato et al., 2007; Zhang et al., 2012; Tong et al., 2013; Li et al., 2017a). However, the primary magnetic components of the Eocene volcanic rocks in the TCT, and the early Miocene secondary magnetic components in the northern edge of the STB, have suggested that the TCT and STB experienced only minor southward extrusion relative to East Asia, since the Oligocene (Kornfeld et al., 2014; Tong et al., 2016). Thus, the process of southward crustal extrusion in the southeastern edge of the Tibetan Plateau may be more complex than previously believed. In fact, the amount of latitudinal crustal shortening in southern Eurasia estimated from paleomagnetic studies is still highly controversial, ranging from 2400 km to less than 200 km (Sun et al., 2010; Tan et al., 2010; Ma et al., 2014; Li et al., 2017b). This may be the result of inclination shallowing of the Cretaceous and Paleogene red-beds (Tauxe and Kent, 2004; Huang et al., 2013) and possible remagnetization of the Cretaceous and Paleogene volcanic rocks of the Lhasa Terrane (LT) (Huang et al., 2015). In addition, the amount of post-collision latitudinal crustal shortening across southern Eurasia was mainly estimated from the Cretaceous and Paleogene paleomagnetic data from the central part of the LT. Therefore, it is unreasonable to evaluate the contribution of southward crustal extrusion in the southeastern part of the Tibetan Plateau to the post-collision crustal shortening across southern Eurasia, based solely on current paleomagnetic studies in the central part of the Tibetan Plateau.

The eastern end of the QT directly faces the northeastern frontend of the EHS (Fig. 1A). A series of Paleogene basins are distributed in the region, for example the Nangqian Basin, Dingqing Basin, Gonjo Basin and Basu Basin. These basins were filled by Paleogene red-beds occasionally interbedded with volcanic rocks. In this study, paleomagnetic analyses and multiple inclination shallowing tests were carried out on the Paleogene red-beds of the Gonjo Basin, at the eastern end of the QT, to reveal the process of southward crustal extrusion in the southeastern edge of the Tibetan Plateau, and its contribution to post-collision crustal shortening in southern Eurasia.

2. Regional geology and sampling

The QT is a narrow E–W extending block, which lies between the LT and the Songpan-Ganzi Terrane (SGT). The Bangong–Nujiang Suture Zone (BNSZ) and Jinshajiang Suture Zone (JSSZ) comprise the southern and northern boundaries of the QT, respectively (Fig. 1A). The Longmo Co-Shuanghu Suture further divides the QT into the Northern and Southern Qiangtang Sub-terranes (NQT and SQT) (Pan et al., 2004; Zhang et al., 2006; Liu et al., 2011; Zhu et al., 2013). The SQT began to assemble onto the NQT since the middle Triassic (244–237 Ma), finally comprising the integrated QT (Yan et al., 2016). Subsequently, since the late Triassic, the QT accreted onto the SGT along the JSSZ (Pullen et al., 2008; Zhu et al., 2013; Yan et al., 2016). In the eastern part of the QT, the tectonic lines composed of faults and fold-axis change gradually from E-W trending in the central Tibetan Plateau to SE-NW trending in the southeastern part of the Tibetan Plateau, and then extend into Yunnan Province with an approximate S-N trending (Fig. 1).

The Gonjo Basin is situated in the eastern part of the QT. It is elongated and extends in a SE-NW direction with a length of ~258 km and a width of ~1.5–18 km (Li et al., 2004) (Fig. 1A, B). The Eocene red-beds strata of the Gonjo Basin are designated the Gonjo Formation in the lower segment and the Ranmugou Formation in the upper segment; they have a total thickness of ~2700 m and are mainly composed of purple-red fine-grained sandstone, siltstones, mudstone, and variegated sandy conglomerate (BGMRX, 1993; Zhou et al., 2003; Li et al., 2004; Studnicki-Gizbert et al., 2008). The Gonjo Formation unconformably overlies Lower and Middle Triassic limestone and volcanic rocks, and there is a conformable contact with overlying Ranmugou Formation. A serious of folds with SE-NW trending fold-axis is developed in the Gonjo

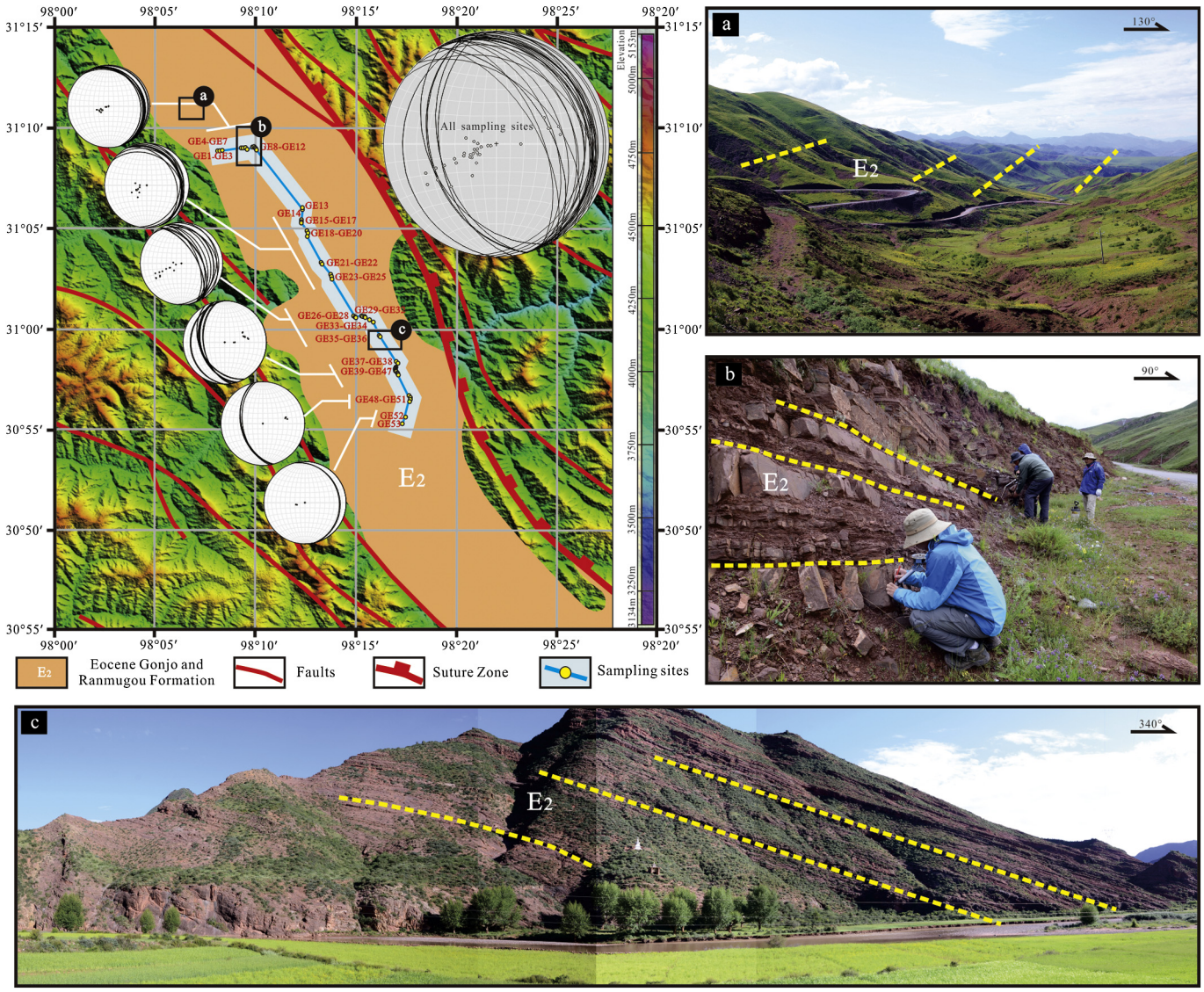


Fig. 2. The sampling section and sampling sites distribution of Gonjo Basia. The projection of measured bedding poles of each sampling site is shown. The position of photographs were marked in figure using a, b and c.

and Ranmugou Formation (Li et al., 2006; Zhu et al., 2006). Abundant Eocene sporopollen fossils and Paleocene guide fossils (palm plants: *Sabalites* sp.) were found in the Gonjo and Ranmugou Formation (BGMRX, 1993). In the northern part of the Gonjo Basin, volcanic rock with the age of 44.2 ± 0.9 Ma (K–Ar dating, Li et al., 2004) and 43.2 ± 0.2 Ma (U–Pb dating, Tang et al., 2017) conformably overlies the Ranmugou Formation. In addition, potassic dykes intruding in the Gonjo Formation in the Nangqian Basin gave an isotopic geochronological age of 37.1–37.8 Ma (Zhu et al., 2006). Overall, the geological evidence indicates that the Gonjo and Ranmugou Formations were formed in the Eocene.

The sampling section was distributed in the two limbs of a fold. A total of 53 sampling sites (584 samples) were drilled from fine-grained sandstone, siltstones and mudstone of the Gonjo and Ranmugou Formations using a portable drill and were oriented with a magnetic compass. Sites GE1–GE36 and GE52–GE53 were collected from the southwestern limb of the fold, and sites GE37–GE51 were collected from the northeastern limb of the fold (Fig. 2). To test the occurrence of depositional and/or compaction-induced inclination shallowing of the red-beds, 8 hand specimens oriented on the strata bedding were obtained from 8 sampling sites. These samples are designated by the letter 'S' in the sample code (e.g., GES2 for site GE2; GES16 for site GE16, etc.).

The location of each sampling site was determined by portable GPS. The present geomagnetic field direction of the sampling section was calculated by the International Geomagnetic Reference Field (2010) (International Association of Geomagnetism and Aeronomy, 2010). All core samples were cut into standard specimens with a length of 2.2–2.3 cm in the laboratory for rock magnetic and thermal demagnetization experiments.

3. Methods

3.1. Rock magnetic measurements and results

Six representative specimens (GE1-4, GE19-3, GE23-9, GE32-12, GE46-2 and GE52-1) were selected for rock magnetic analysis, based on lithological characteristics. The progressive acquisition of an isothermal remanent magnetization (IRM) was used to reveal the coercivity spectra of the magnetic minerals. Thermal demagnetization of a three-components IRM (applying successively to each of the three axes with DC field of 2.5 T for Z axis, 0.4 T for X axis,

and 0.12 T for Y axis) was used to reveal the unblocking temperature of the magnetic minerals (Lowrie, 1990).

Six specimens exhibit consistent rock magnetic results (Supplementary Fig. 1). In these specimens, the IRM acquisition curves show that the remanence increased slowly and did not approach saturation when the applied DC field was gradually increased from 0 to 2500 mT. In the process of applying reversed DC fields, the remanence reduced to zero in the range of 600–750 mT, indicating the presence of the high coercivity magnetic mineral hematite (Supplementary Fig. 1A). Thermal demagnetization of the three-components of these six specimens show that the hard and intermediate magnetization components decreased significantly at 80 °C, indicating the presence of goethite. The hard and intermediate components decreased slightly when the thermal demagnetization temperature reached 350–400 °C, indicating the possible presence of pyrrhotite or maghemite. When the thermal demagnetization temperature reached 580 °C, the soft components decreased to zero, and the hard and intermediate components exhibited inflexions in the temperature interval of 560–580 °C, indicating the presence of magnetite. When the thermal demagnetization temperature exceeded 580 °C, the hard and intermediate components decreased rapidly and were finally unblocked at 680 °C, indicating an abundance of hematite (Supplementary Fig. 1B). Thus, hematite and magnetite are the main magnetic carriers in the red-beds of the Gonjo and Ranmugou Formations, and a small amount of pyrrhotite or maghemite, together with goethite, is also present.

3.2. Paleomagnetic measurements and results

All specimens were subjected to stepwise thermal demagnetization using an ASC TD48 thermal demagnetizer, and the remanence after each step was measured using a 2G-755 cryogenic magnetometer, in the Key Laboratory of Paleomagnetism and Tectonic Reconstruction, the Ministry of Land and Resources in Beijing. From the rock magnetic results, the stepwise thermal demagnetization was performed up to 685 °C, with an initial temperature interval of 50–100 °C and then with an interval of 10–20 °C, at higher temperatures. The results were plotted on Zijderveld diagrams (Zijderveld, 1967). The directions of the magnetic component were determined using principal components analysis (Kirschvink, 1980) for most of the specimens, while the best-fit great circles method (McFadden and McElhinny, 1988) was used in a small number of cases. The mean directions of the sampling section were calculated using Fisher statistics (Fisher, 1953).

The low-temperature magnetic components (LTC) from 303 specimens were isolated in the temperature interval of NRM (20–300 °C (Fig. 3). The sample-mean direction of the LTC is $D_g = 355.1°$, $I_g = 46.9°$, $k = 18.7$, $\alpha_{95} = 1.9°$, $n = 303$ (samples) before tilt correction, and $D_s = 351.7°$, $I_s = 37.1°$, $k = 6.1$, $\alpha_{95} = 3.6°$ after tilt correction (Fig. 4A). The sample-mean direction before tilt correction is close to the present geomagnetic field direction ($D = 4.3°$, $I = 48.7°$) and the geocentric axial dipole direction ($D = 0°$, $I = 51.7°$) of the sampling area. The precision parameter (k) decreases from 18.7 (k_g) to 6.1 (k_s) after tilt correction, indicating that the LTC is a viscous remanent magnetization.

In 61 specimens, middle-temperature magnetic components (MTC) were isolated in the temperature interval of 300 °C ∼ 580 °C, indicating that the magnetic carriers are magnetite and pyrrhotite/maghemite (Fig. 3). The sample-mean direction of the MTC is $D_g = 359.7°$, $I_g = 42.8°$, $k = 16.4$, $\alpha_{95} = 4.6°$, $n = 61$ (samples) before tilt correction, and $D_s = 353.7°$, $I_s = 32.0°$, $k = 5.8$, $\alpha_{95} = 8.3°$ after tilt correction (Fig. 4B). The precision parameter (k) decreases from 16.4 (k_g) to 5.8 (k_s) after tilt correction, suggesting that the MTC is a post-folding original magnetic component.

Except for sites GE12, GE35 and GE48, in the specimens from the other 50 sampling sites high-temperature magnetic components (HTC) could be isolated. For most specimens, the thermal demagnetization results indicate a linear decay of the magnetization towards the origin, and the HTC can be isolated within the temperature interval of 580–685 °C, using principal components analysis (Kirschvink, 1980) (Fig. 3A). The magnetic component vectors of 48 specimens did not trend towards the origin when the thermal demagnetization temperature exceeded 600 °C; however, they changed along great circles on equal-area projections. In these specimens, the best-fit great circles and intersection of great circles methods (McFadden and McElhinny, 1988) were used to isolate the HTC (Fig. 3B). Finally, 36 sampling sites exhibited reversed polarity, and the other 14 sampling sites exhibited normal polarity (Supplementary Table 1). If the sites GE9-GE11, GE13, GE15, GE18, GE25-29, GE43, GE49 and GE51-53 (with $\alpha_{95} > 15°$) are excluded from the Fisher's statistic, the resulting site-mean direction of the HTC is $D_g = 30.3°$, $I_g = 31.5°$, $k = 8.9$, $\alpha_{95} = 8.6°$, $N = 34$ (sampling sites) before tilt correction, and $D_s = 33.6°$, $I_s = 28.6°$, $k = 39.7$, $\alpha_{95} = 3.9°$ after tilt correction (Supplementary Table 1). However, the site mean direction with $\alpha_{95} > 15°$ was used to calculate the site-mean direction in some previous paleomagnetic studies (e.g., $\alpha_{95} < 29.9°$, Otofuji et al., 2007; $\alpha_{95} < 26.2°$, Sun et al., 2012; $\alpha_{95} < 21.5°$, Achache et al., 1984; $\alpha_{95} < 18.7°$, Tan et al., 2010). Thus, if the sampling sites with $\alpha_{95} < 25°$ are all used for Fisher's statistic, then the resulting site-mean direction of the HTC is $D_g = 31.3°$, $I_g = 34.0°$, $k = 9.0$, $\alpha_{95} = 7.7°$, $N = 43$ (sampling sites) before tilt correction, and $D_s = 35.5°$, $I_s = 29.3°$, $k = 45.9$, $\alpha_{95} = 3.2°$ after tilt correction (Supplementary Table 1, Fig. 4C), which is consistent with that calculated using the sampling sites with $\alpha_{95} < 15°$. The fold test of McElhinny (1964) shows that $k_s/k_g = 5.08 > F[84, 84] = 1.43$ at the 95% confidence limit, indicating a positive fold test at that level. In the McFadden fold test (McFadden, 1990), the calculated values are $\xi_{(2)\text{in situ}} = 17.69$ in geographic coordinates and $\xi_{(2)\text{tilt corrected}} = 1.37$ after tilt correction, while the critical value is $\xi_C = 7.54$ at the 95% confidence limit, also indicating a positive fold test at that level. The reversal test (McFadden and Lowes, 1981; McFadden and McElhinny, 1990) is positive at the 95% confidence level in classification B, with statistical parameters $R = 7.4$, $F[16, 64] = 2.11$, and critical Gamma = 7.9. In addition, the precision parameter (k) reaches a maximum ($k_{max} = 48.86$) at 89.3% unfolding during progressive unfolding (Watson and Enkin, 1993), and the direction ($D_m = 35.5°$, $I_m = 29.9°$, $k = 48.86$, $\alpha_{95} = 3.1°$) is consistent with the site-mean direction after tilt correction (Fig. 4C). Thus, the mean direction of the HTC after tilt correction is a primary magnetic component. The paleopole of each sampling site was calculated, and the average paleopole (54.3°N/204.6°E, $A_{95} = 2.9°$) of the Gonjo and Ranmugou Formations was calculated using Fisher's statistic (Fisher, 1953) (Supplementary Table 1).

Previous isotopic geochronological studies indicated that the volcanic rocks at the top of the Eocene strata in the northern part of the Gonjo Basin were formed at 44.2 ± 0.9 Ma (K–Ar dating, Li et al., 2004) and 43.2 ± 0.2 Ma (U–Pb dating, Tang et al., 2017). In addition, sedimentological studies of the Nangqian Basin and Gonjo Basin indicate that several thick, variegated, sandy conglomerate layers occur in the lower and middle segments of the Gonjo Formation (Zhou et al., 2003). In the lower part of our sampled section, two thick sandy conglomerate layers occur between sites GE3 and GE4, and GE12 and GE20, indicating that the sampled section spans the lower segment of the Gonjo Formation. Thus, the sampled strata should approximately span the time interval of 56.0–43.2 Ma, indicating that the primary magnetic components were acquired during this period.

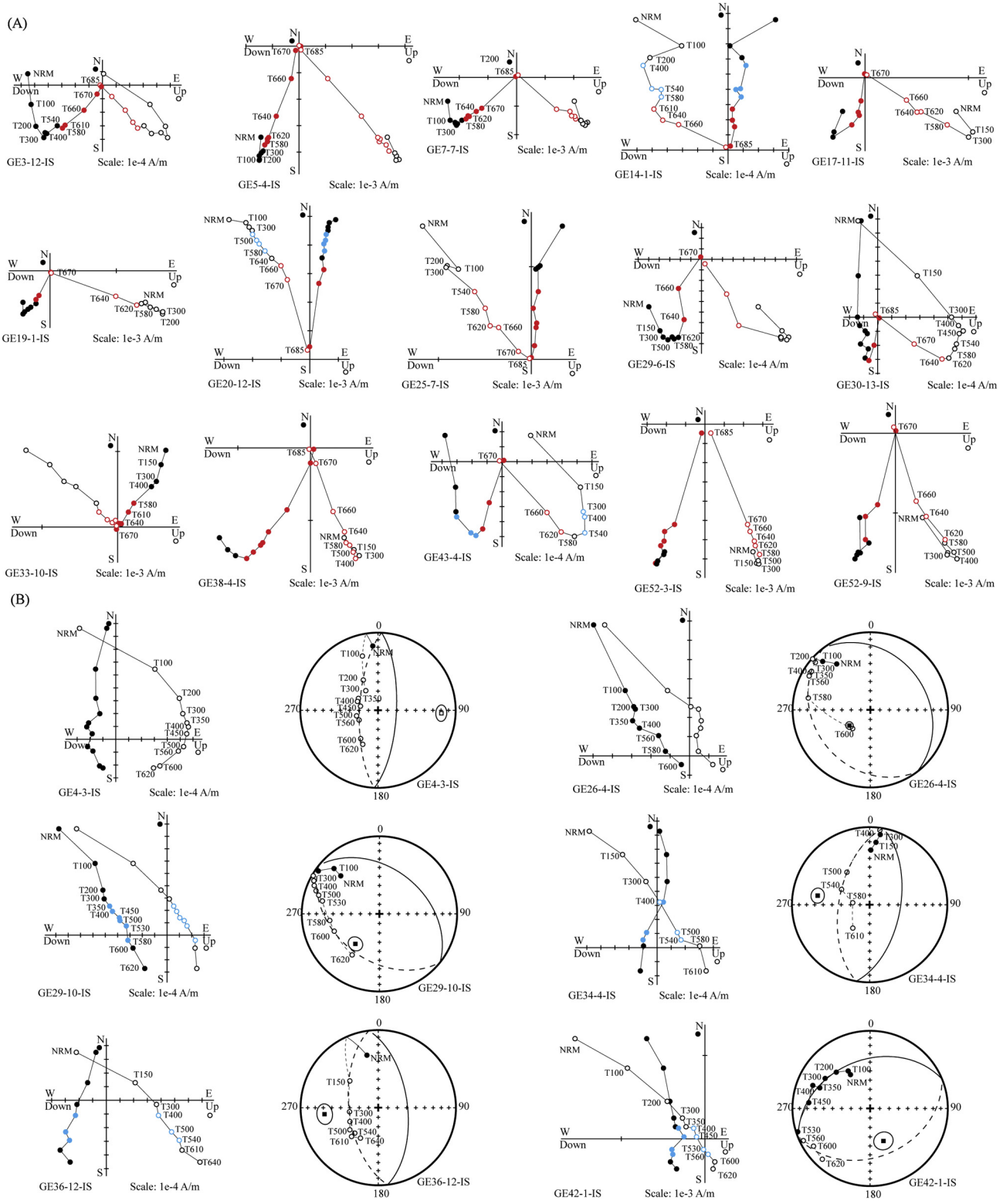


Fig. 3. Orthogonal projections of magnetization vector end-points for representative specimens of the Eocene Conjo and Ranmugou Formations in geographic coordinates. (A): High-temperature magnetic components were isolated using the principal components analysis (Kirschvink, 1980) in temperature interval of 580 °C ~ 685 °C. (B) High-temperature magnetic components were isolated using the best-fit great circles and intersection of great circles method (McFadden and McElhinny, 1988). Solid/open symbols are the horizontal/vertical projection in geographic coordinates. The red/blue points show the thermal temperature intervals used for isolating the HTCs/MTCs. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4. Corrections for inclination shallowing

Because of potential depositional and compaction-induced inclination shallowing of the red-beds in East Asia, it is very important to carry out inclination shallowing correction of the paleomagnetic data of the Cretaceous and Paleogene red-beds (Tauxe and Kent, 2004). The anisotropy-based inclination shallowing correction method of Hodych and Buchan (1994) and the elongation/incli-

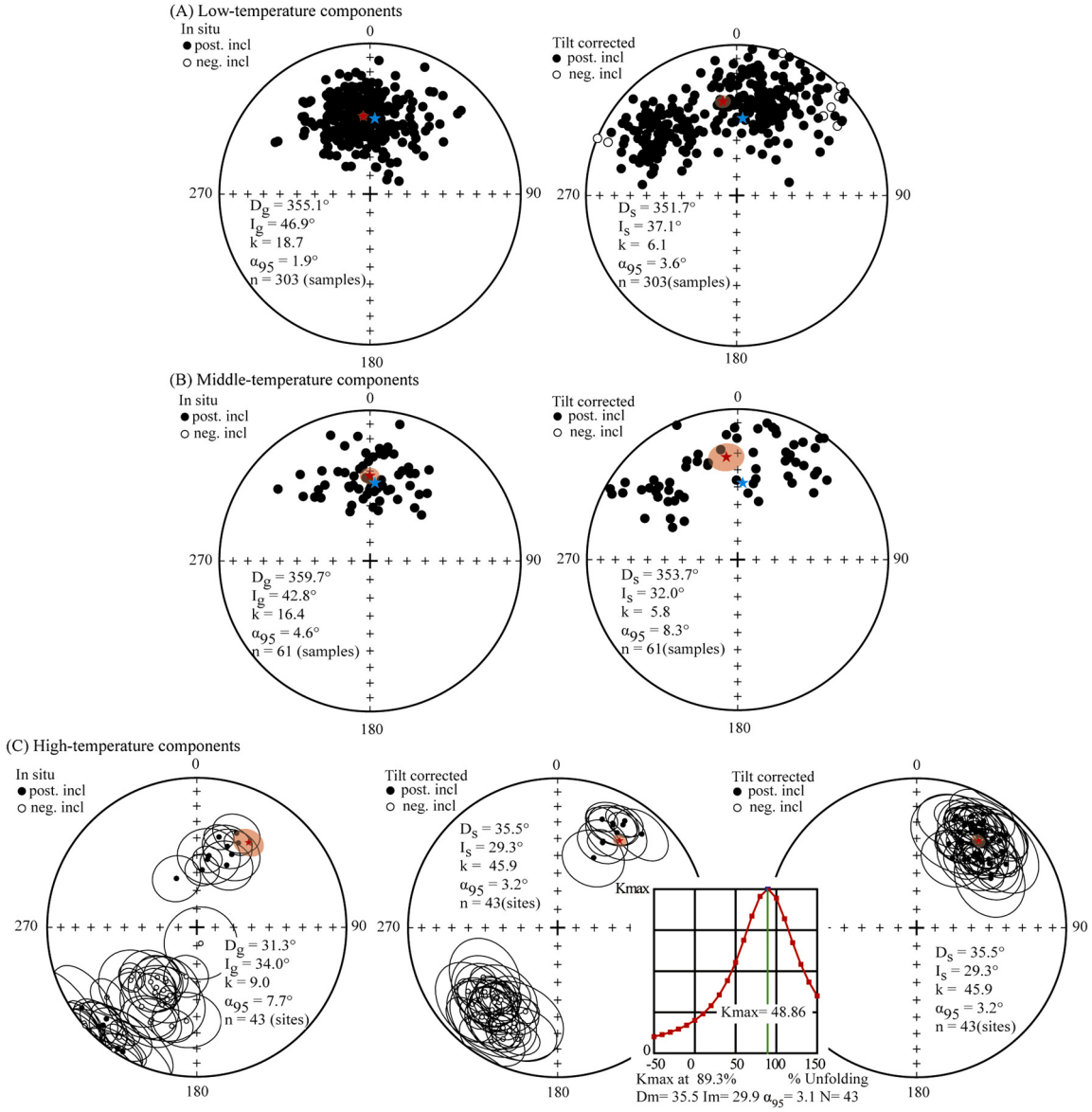


Fig. 4. (A): Equal-area projections of the sample-mean direction for low-temperature magnetic components before and after tilt correction. (B): Equal-area projections of the sample-mean direction for middle-temperature magnetic components before and after tilt correction. (C): Equal-area projections of site-mean direction for high-temperature magnetic components. Red shaded circles around the red stars in the (A), (B) and (C) denote the 95% confidence limit and mean-directions of magnetic components. The blue stars in the (A) and (B) denote the present geomagnetic direction of sampling section. Solid and open symbols denote the lower and upper hemisphere before and after tilt correction, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

nation (E/I) correction method of Tauxe and Kent (2004) are two commonly used methods and were applied in the present study.

Eight hand samples from 8 sampling sites oriented on the strata bedding plane were used for anisotropy-based inclination shallowing correction**.** A total 32 standard specimens were drilled from the 8 hand samples in the direction perpendicular to the strata bedding. These specimens were subjected to successively increasing DC fields (in the range of 0–1000 mT) applied at a 45° angel to the sampling strata bedding plane. In the process of applying the DC fields, the IRM parallel (IRMx) and perpendicular (IRMz) to the bedding plane of each specimen was measured after each application of the DC field. Subsequently, thermal demagnetization (in the temperature range of 100–695 °C) was also performed on these specimens (Supplementary Fig. 2). The values of IRMz/IRMx of 32 specimens in the range of 200–1000 mT changed from 0.6355 to 0.8905. The values of IRMz/IRMx in the thermal demagnetization temperature interval of 150–685 °C and above 580 °C changed from 0.6419 to 0.8884 and from 0.6271 to 0.8978, respectively (Supplementary Table 2). Given that the HTC was within 580–685 °C, the mean IRMz/IRMx of each hand sample in the temperature range of 580–685 °C was used to calculate the mean IRMz/IRMx of the sampling section (0.7730 ± 0.0603) (Supplementary Table 2). The site-mean inclination of the Gonjo Formation increased from 29.3° to 41.6° after correction for inclination shallowing.

Previous research has indicated that intense crustal deformation and local rotational deformation with a vertical-axis could lead to excessive inclination correction when using the E/I correction method of Tauxe and Kent (2004) (Tong et al., 2015). In this study, the strata bedding of two limbs of the fold remain relatively stable and the amount of specimens in each limb of the fold exceeds 100. Thus, the E/I correction can be performed on the paleomagnetic data from the northeastern and southwestern limbs of the fold, respectively (Supplementary Fig. 3). The site-mean direction of the

sampling sites in the southwestern limb of the fold (GE1–GE8, GE10–GE12, GE14–GE25, GE28–GE34 and GE36) is $D_s = 34.1°$, $I_s = 27.8°$, $k = 34.2$, $\alpha_{95} = 4.6°$ after tilt correction. The corresponding E/I corrected inclination increased to 40.3°, with an error range of 34.0–45.3° (Supplementary Fig. 3A). The site-mean direction of the sampling sites in the northeastern limb of the fold (GE37–GE47, GE50, GE51) is $D_s = 35.0°$, $I_s = 31.8°$, $k = 66.9$, $\alpha_{95} = 7.3°$ after tilt correction. The corresponding E/I corrected inclination increased to 40.4°, with the error range of 33.5–46.7° (Supplementary Fig. 3B). The E/I correction results are consistent with the anisotropy-based inclination shallowing correction result. In the following discussion, the measured site-mean inclination of the Gonjo Formation was replaced by the anisotropy-based inclination shallowing corrected inclination (Hodych and Buchan, 1994), which gave a paleopole of 57.9°N/192.1°E, $A_{95} = 2.9°$.

5. Discussion

5.1. Post-collision crustal shortening in southern Eurasia

The Paleocene collision of India and Eurasia induced enormous latitudinal crustal shortening across southern Eurasia (Tapponnier et al., 1982; Meyer et al., 1998; Yin and Harrison, 2000; DeCelles et al., 2002; Spurlin et al., 2005; Yin et al., 2007; Yang et al., 2014; Li et al., 2017b). In this study, Paleocene and Eocene paleomagnetic data of the LT were compared with that of the QT to estimate the crustal shortening across Tibet (Table 1). To ensure their reliability, the Paleogene paleomagnetic data for LT and QT were first filtered according to the known sources of inclinational divergence, such as depositional and/or compaction-induced inclination shallowing of red-beds (Tauxe, 2005) and the remagnetization of volcanic rocks (Huang et al., 2015). The pre-folding original remanence direction of Halim et al. (1998), obtained from the Paleocene red-beds on the northern edge of the QT was not subjected to any inclination shallowing tests. Tan et al. (2010) obtained a pre-folding remanence direction from the Eocene tuffs in the central part of the LT, however, the absence of detailed rock magnetic analysis and the single polarity of their data made it difficult to determine the acquisition time of the pre-folding remanence. Thus, these two paleomagnetic data were excluded from the subsequent discussion. The 60 Ma, 50 Ma, and 40 Ma poles for Eurasia (Torsvik et al., 2008) are almost identical (Table 1), indicating that the Eurasia remained almost stationary during the interval from 60–40 Ma. Thus, the 60–40 Ma poles for Eurasia were selected as reference poles.

Detailed comparison shows that the paleolatitude of the LT exhibits a systematic pattern of variation, moving from lower northern latitude to higher northern latitude from 64–43 Ma (Fig. 5). The LT was situated at the paleolatitude of 7.5 ± 8.5°N from 64–60 Ma. It then moved to 13.7 ± 7.3°N at 55 Ma, and to 16.5 ± 5.7°N at 53 Ma. Subsequently, the LT moved to the paleolatitude of ~22°N (at the paleolatitude of 22.3 ± 5.0°N in 54–47 Ma, and/or 22.8 ± 2.6°N in 54–43 Ma) (Fig. 5). The paleolatitudes calculated from the paleomagnetic data of Achache et al. (1984) and Chen et al. (2010) are clearly different to those calculated for the LT, which may indicate that the paleolatitudes are eclectic and inaccurate because of the wide age range of the paleomagnetic data (Fig. 5). Thus, the small-circle fitting method of Mardia and Gadsden (1977) is unsuitable for all the Paleogene paleopoles combined to estimate the paleolatitude of the LT because of the wide age range of paleomagnetic data, which would give inaccurate results.

The paleomagnetic data of Huang et al. (2013) and Dupont-Nivet et al. (2010) were obtained from strata of similar age in the adjacent region, and thus the small circle fitting method of Mardia and Gadsden (1977) can be applied to them. This gave a colatitude of 67.4 ± 0.7°N, centered on the reference point of 30.0°N, 91.0°E, indicating that the central part of the LT was situated at the paleolatitude of 22.6 ± 0.7°N during the interval from 54–43 Ma (Table 1, Fig. 6A). The 60–40 Ma reference poles for Eurasia (Torsvik et al., 2008) were used to calculate the expected paleolatitude of reference point (30.0°N, 91.0°E). The expected paleolatitude is 33.4 ± 2.9°N, indicating that the north of the central LT experienced ~10.8 ± 3.0°(~1250 ± 350km) latitudinal crustal shortening since 54–43 Ma (Fig. 6A). The 55 Ma and 53 Ma paleomagnetic data for the LT show that it was situated at the paleolatitudes of 13.7 ± 7.3°N and 16.5 ± 5.7°N (reference point: 30.0°N, 91.0°E), respectively, indicating that the north of the central LT accumulated 19.7 ± 7.8° (2300 ± 890 km) of crustal shortening since 55 Ma, and 16.9 ± 6.4° (1900 ± 730 km) of crustal convergence since 53 Ma (Fig. 6B).

Three late Eocene paleomagnetic data values for the central part of the QT were selected for consideration (Table 1). The paleomagnetic data from volcanic rocks with the acquisition ages of 38.6 ± 0.5 Ma and 35.4 ± 2.4 Ma gave paleolatitudes of 27.5 ± 7.8°N and 22.8 ± 5.2°N, respectively (Lippert et al., 2011), indicating that the north of the central QT experienced 10.6 ± 6.6° ~ 12.9 ± 7.4° of crustal shortening since ~35 Ma, relative to the 40 Ma reference pole for Eurasia (Torsvik et al., 2008) (Table 1). However, it should be noted that another paleomagnetic result from volcanic rocks with the acquisition age of 34.2 ± 1.3 Ma, in the central QT indicates that the sampling area was situated at the paleolatitude of 34.8 ± 6.6°N. This, indicates that the north of the QT experienced minor crustal shortening (0.5 ± 5.7°N) (Lippert et al., 2011). This controversial result is likely caused by paleosecular variation that was recorded by the volcanic rock. Lippert et al. (2011) suggested that the three sampling sections would have averaged out the paleosecular variation after the data were combined, however, without further confirmation we cannot certain that the paleomagnetic data from volcanic rocks with the age of 34.2 ± 1.3 Ma have averaged out the paleosecular variation. Consequently, this paleomagnetic data value is excluded from the discussion. Because of the similar acquisition time of the two paleomagnetic data values, the small circle fitting method of Mardia and Gadsden (1977) was applied. This gave a colatitude of 67.8 ± 2.1°N, centered on the reference point of 30.0°N, 91.0°E, indicating that the central part of the QT was situated at the paleolatitude of 22.2 ± 2.1°N during the interval between 38.6 ± 0.5 Ma and 35.4 ± 2.4 Ma. Consequently, the north of the central QT accrued ~11.2 ± 3.6° (~1300 ± 410km) of latitudinal crustal shortening relative to the reference pole for Eurasia (Fig. 6C). The Eocene paleomagnetic data from the Gonjo Basin indicated that the eastern part of the QT was situated at the paleolatitude of 23.9 ± 2.9°N during the interval from 56.0–43.2 Ma. The expected paleolatitude (35.5 ± 2.7°N) of the sampling area (31.0°N, 98.2°E) was calculated from the 60–40 Ma reference poles for Eurasia (Torsvik et al., 2008), and indicates that the north of the Gonjo Basin experienced ~11.6 ± 3.9° (~1300 ± 450km) of latitudinal crustal shortening since 56.0–43.2 Ma (Fig. 6D).

5.2. Uplift of the Tibetan Plateau

Compression-induced crustal tectonic deformation and thickening is one of the main causes of the post-collision crustal shortening of southern Eurasia, and it indirectly reflects the uplift of the Tibetan Plateau. Previous geological and paleomagnetic studies have indicated that the E-W extending LT and QT collided along the Bangong–Nujiang Suture Zone since the Middle-Late Jurassic, and since the Late Cretaceous they constituted the stationary southern edge of Eurasia prior to the collision of India and Eurasia (Dewey et al., 1988; Murphy et al., 2000; Metcalfe, 1996, 2006; Kapp et al., 2007; Chen et al., 2010; Zhu et al., 2011; Yan et al., 2016). The process of Paleogene paleolatitudinal change

Table 1
The Paleogene paleomagnetic data from the Lhasa Terrane and Qiangtang Terrane.

Site (°N/°E)	Time (Ma)	*N*	Observed			Paleopole			Paleolat. (°)	Translation (°)	Relative to	Rock type	Fold test	Inclination shallowing correction	Magnetic carrier	Reference
			Dec. (°)	Inc. (°)	α_{95} (°)	Plat. (°)	Plon. (°)	A_{95} (dp/dm) (°)								
Qiangtang Terrane																
#34.5/92.8	P	7	9.5	34.3	8.9	72.3	242.0	5.9/10.2	–	–	–	R	P	N	–	Halim et al., 1998
#32.2/86.6	E(34.2 ± 1.3)	7	106.5	−54.1	3.7	30.0	21.1	6.6	34.8 ± 6.6	0.5 ± 5.7	C	V	P	–	magnetite	Lippert et al., 2011
32.7/85.6	E(35.4 ± 2.4)	9	15.5	40.0	7.2	73.1	207.6	8.8	22.8 ± 5.2	12.9 ± 7.4	C	V	P	–	magnetite	Lippert et al., 2011
34.5/90.2	E(38.6 ± 0.5)	4	355.8	46.1	3.4	82.1	298.4	7.8	27.5 ± 7.8	10.6 ± 6.6	C	La	P	–	magnetite	Lippert et al., 2011
Small circle fitting results: Reference point: 30.0°N/91.0°E Paleolatitude = 22.2 ± 2.1°N Expected paleolatitude = 33.4 ± 2.9°N Crustal shortening = 11.2° ± 3.6° (∼1300 ± 410 km) Relative to the 60–40 Ma reference poles of Eurasia																
31.0/98.2	E(57.2–47.3)	43	35.5	41.6	3.2	57.9	192.1	2.9	23.9 ± 2.9	11.6 ± 3.9	A + B + C	R	P + D	E + A	–	This study
Lhasa Terrane																
#30.0/91.2	E (43–40)	9	359.5	51.8	5.2	87.5	81.4	5.9	–	–	–	T	P	–	No magnetic rock analysis	Tan et al., 2010
30.0/91.2	E (54–43)	–	10.2	40.0	2.6	78.4	217.2	2.6	22.8 ± 2.6	11.6 ± 2.9	B	R + T	M	E + A	magnetite for tuff	Huang et al., 2013
30.0/91.1	E (54–47)	24	12.5	39.4	5.3	76.4	212.6	5.0	22.3 ± 5.0	12.1 ± 4.5	B	V	P	–	magnetite	Dupont-Nivet et al., 2010
Small circle fitting results: Reference point: 30.0°N/91.0°E Paleolatitude = 22.6 ± 0.7°N Expected paleolatitude = 33.4 ± 2.9°N Crustal shortening = 10.8° ± 3.0° (∼1200 ± 350 km) Relative to the 60–40 Ma reference poles of Eurasia																
30.0/91.1	E(53)	9	13.4	30.8	6.3	71.8	225.7	5.7	16.5 ± 5.7	16.9 ± 6.4	A + B + C	D	P + D	–	titanomagnetite	Liebke et al., 2010
30.1/90.9	P(55)	14	359.0	26.1	9.2	73.6	274.3	7.3	13.7 ± 7.3	19.7 ± 7.8	A + B + C	T	P	–	magnetite	Sun et al., 2010
30.0/91.2	P(64–60)	15	353.5	14.8	8.8	66.0	284.9	8.5	7.5 ± 8.5	25.9 ± 9.0	A + B + C	A + T	P + D		Magnetite and hematite	Chen et al., 2010
#30.0/91.2	P–E (60–44)	23	355.9	20.2	6.9	70.6	283.1	5.3	10.5 ± 5.3	–	–	R + T + A + Li	P	–	magnetite for tuff and adesite	Chen et al., 2010
#30.0/91.0	E(60–45)	8	350.9	25.5	11.0	71.4	299.8	11.0	13.4 ± 6.4	–	–	V	D	–	magnetite	Achache et al., 1984
APWP of Eurasia	A: 60					79.0	166.8	2.4								Torsvik et al., 2008
	B: 50					79.1	154.2	2.6								
	C: 40					82.3	150.5	2.8								

N is the number of sampling sites used for paleomagnetic calculation. Dec. and Inc. are declination and inclination, respectively. Plat. and Plon. are latitude and longitude of paleopoles. α_{95} and A_{95} is the radius of cone at 95 percent confidence level about the mean direction. dp/dm is the confidence limits of poles. Paleolat. is the calculated paleolatitude, positive means the north latitude, negative means the south latitude. # means the paleomagnetic data were not used in the discussion. Time (Ma): P, Paleocene. E, Eocene. Rock type: R. Red-beds; V, volcanic rock; La, lava flow; T, tuff; D, dyke; A, Andesite; Li, limestone. Fold Test: P, positive fold test; M, the A_{95} value of the mean VGP of sampling section falls just within the range of A_{95} values defined as a reliability envelope by Deenen et al. (2011); D, coexistence of dual-polarity. Inclination Shallowing correction: N, not applied inclination shallowing correction; E, E/I correction; A, anisotropy-based inclination correction.

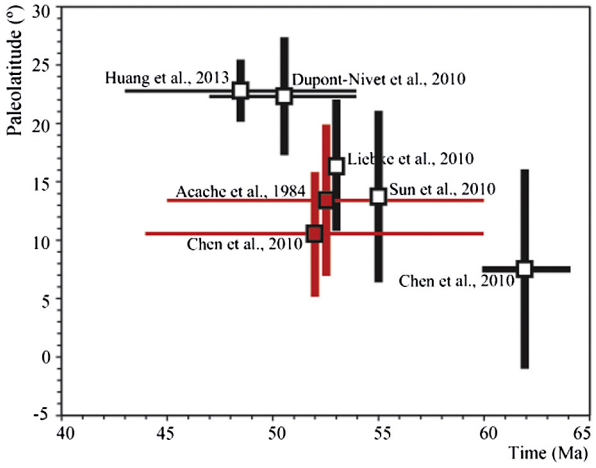


Fig. 5. The paleolatitudinal systematic variation of the central part of LT in the Paleogene. The LT gradually moved from lower latitude to high latitude from 64–43 Ma. The black points denote the paleomagnetic data were used to constrain the paleolatitudinal variation of the LT. The red points denote the paleomagnetic data were excluded from the discussion because of the large uncertain of age. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

of the LT and QT indicates that the north of the E-W extending LT and QT experienced significant crustal shortening (1250 ± 350 km–1300 ± 450 km) relative to the reference poles of Eurasia, since the period of 56.0–43.2 Ma. The central part of the QT also experienced a similar amount of crustal shortening (∼1300 ± 410 km) relative to the reference pole of Eurasia sine the period of 38.6 ± 0.5 Ma and 35.4 ± 2.4 Ma. It can be concluded that the north of the QT remained roughly stationary before 35.4 ± 2.4 Ma, and that subsequently it experienced significant crustal shortening.

Recent paleomagnetic results suggested that the initial collision of the Indian Plateau and Eurasia occurred in the late Paleocene (Ma et al., 2014; Yang et al., 2014; Li et al., 2017b). Thus, the 64–60 Ma paleolatitude of the central part of the LT (7.5 ± 8.5°N) is suitable for representing the position of the southern edge of Eurasia prior to the collision. This indicates that southern Eurasia experienced a total of ∼2900 ± 1000 km post-collision crustal shortening (Fig. 7B), within this total, ∼ 1600 ± 1100 km of crustal shortening occurred in the LT and QT prior to 54–43 Ma, and the remaining 1300 ± 450 km of latitudinal crustal shortening occurred in the north of the LT and QT since 35.4 ± 2.4 Ma. The large error for the 64–60 Ma paleolatitude of the central part of the LT (Chen et al., 2010) makes it difficult to precisely determine the true amount of crustal shortening in the LT and QT. Nevertheless, it still demonstrates that the LT and QT experienced significant latitudinal crustal shortening before 54–43 Ma.

The foregoing analysis of the post-collision crustal shortening of southern Eurasia demonstrates the LT and QT experienced crustal shortening before 54–43 Ma, while the north of the QT remained stationary during this interval. Subsequently, since 35.4 ± 2.4 Ma, the north of the QT began to undergo significant crustal shortening. In the southeastern edge of the Tibetan Plateau, the Gaoligong strike-slip fault and the Ailao-Red River strike-slip fault comprised the western and northeastern crustal extrusion boundaries, respectively. Isotope chronological studies indicate that they simultaneously commenced strike-slip movement at ∼33–32 Ma (Leloup et al., 1995; Wang and Burchfiel, 1997; Wang et al., 2001; Gilley et al., 2003; Akciz et al., 2008; Wang et al., 2008), which indicates that the southward crustal extrusion in the southeastern edge of Tibet occurred no earlier than the Oligocene. Thus, crustal shortening of the LT and QT began before 54–43 Ma was irrelevant for the southward crustal extrusion, and was mainly induced by crustal tectonic deformation and thickening. This crustal shortening process indicates that southern and central Tibet probably experienced initial uplift in the Early Eocene, and that subsequently, the north of the QT began to uplift since the late Eocene (35.4 ± 2.4 Ma). The deep seismic reflection profile across the Yarlung Zangbo Suture shows that the deepest part of the Indian crust is underthrust just to the north of the YZS (Gao et al., 2016). This indicates that the crustal thrusting and folding movement of the interior of the Tibetan Plateau was probably the principal cause of the uplifting of the Plateau.

Numerous geological studies also suggest that southern Tibet had already been uplifted in the Eocene. The reconstructed paleoelevations in the Himalaya and the LT show that the southern Tibetan Plateau had already been uplifted to near-modern altitudes in the late Oligocene (Currie et al., 2005; Mulch and Chamberlain, 2006; Polissar et al., 2009; Xu et al., 2010; DeCelles et al., 2011); and that the LT, in particular, had been uplifted to an elevation of more than 4000 m before the Oligocene (Rowley and Currie, 2006; DeCelles et al., 2007). Xu et al. (2013) suggests that the QT had been uplifted to the elevation of about 5000 m during the middle Eocene and subsequently remained at a relatively constant elevation from the Oligocene to the present, based on oxygen isotope records from fluvial/lacustrine carbonate cements, pedogenic carbonates and lacustrine marls from the northern QT. Isotope chronology studies of the widely distributed north–south trending diabase and andesitic porphyry dikes in central-southern Tibet show that they were mainly emplaced in the Eocene (47∼38 Ma), which implies that Eocene east–west extension was widespread across the LT and QT, and that the southern Tibetan Plateau had already attained a near-maximum elevation in the Eocene (Lan et al., 2007; Yang et al., 2008; Wang et al., 2010). In the southeastern edge of the Tibetan Plateau, the movement of the Baoshan Terrane and the TCT was gradually transformed from eastward extension into clockwise rotation, indicating that the crustal shortening and thickening in the eastern part of the Tibetan Plateau had reached an extreme in the late Oligocene (Tong et al., 2016). Tectonic and basin analysis studies show that the crust of the central and southern Tibetan Plateau began to thicken since the Eocene (Spurlin et al., 2005; Aikman et al., 2008); and low temperature thermochronology data suggest that the southwestern Tibetan Plateau was uplifted by at least 35 Ma (Van der Beek et al., 2009).

5.3. Southward crustal extrusion in the eastern edge of the Tibetan Plateau

The current southeastern edge of the Tibetan Plateau is composed of the TCT, the STB and the ICB. The LT and QT extend from central Tibet eastwards, taper off to the southeast at the EHS, and then are gradually superseded by the STB. However, the boundaries between the QT, LT and the STB are poorly defined around the EHS. Recent geological studies suggest that the TCT was the eastern extension of the LT, from similarities of the spatiotemporal distribution of igneous rocks and magmatic compositions observed in the TCT and LT (Zhang et al., 2013; Liao et al., 2015; Xie et al., 2016) (Fig. 7). The STB was suggested to be the eastern extension of the SQT (Zhu et al., 2009a, 2009b; Cao et al., 2014; Xie et al., 2016) (Fig. 7).

Previous geological studies suggested that a significant amount of post-collision latitudinal crustal shortening across southern Eurasia was accommodated by the southward extrusion of the TCT, STB and ICB (Tapponnier et al., 1990; Leloup et al., 1995; Lacassin et al., 1997; Royden et al., 2008; Clark and Royden, 2000; Wang and Burchfiel, 2000; Yin and Harrison, 2000; Leloup et al., 2001; Replumaz and Tapponnier, 2003; Molnar and Dayem, 2010). The Red-River strike-slip fault, as the northeastern extrusion boundary, began strike-slip movement since the Oligocene (∼32 Ma), and accumulated a total of 200–800 km left-lateral off-

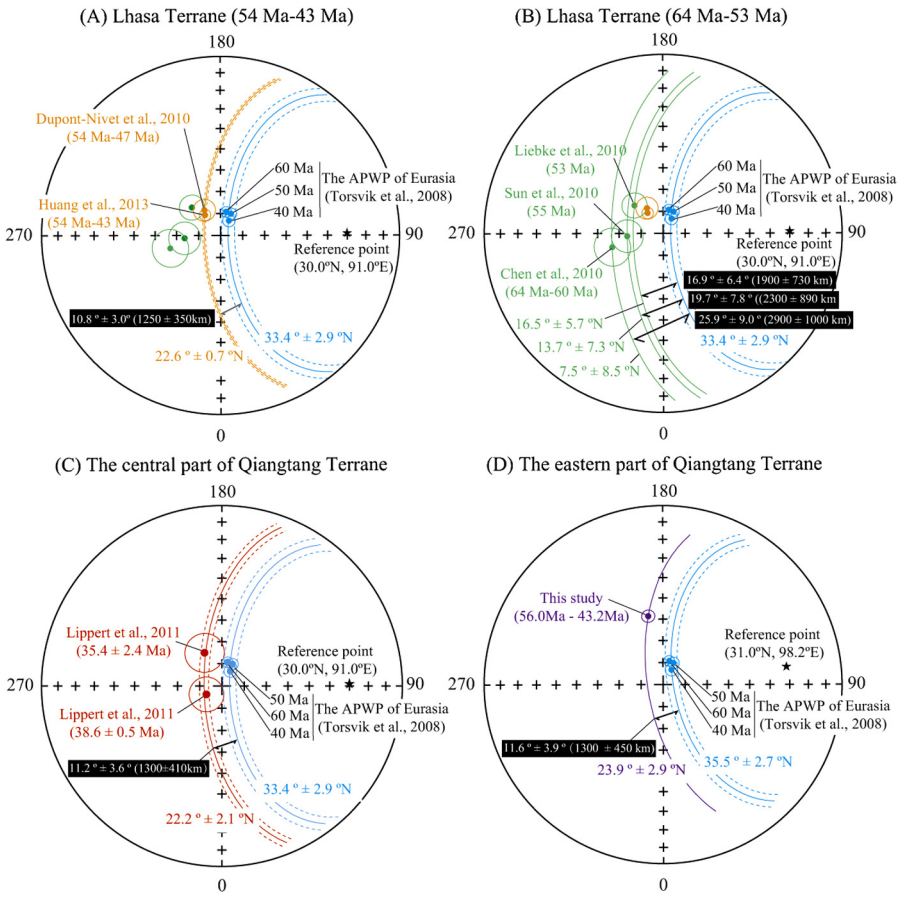


Fig. 6. (A) Equal-area projection of paleopoles of LT with the age of 54–43 Ma. Small-circle fitting passes through 2 selected Paleogene poles of the LT and gives a paleolatitude of 22.6° ± 0.7°N (reference point: 30.0°N, 91.0°E). The expected Paleogene poles of the reference point (30.0°N, 91.0°E) is 33.4 ° ± 2.9°N. (B) Equal-area projection of the 64–53 Ma paleopoles of the LT. The LT was situated at the paleolatitude of 7.5° ± 8.5°N in 64–60 Ma,then moved to the paleolatitude of 13.7° ± 7.3°N at 55 Ma, and finally arrived the paleolatitude of 16.5° ± 5.7°N at 53 Ma. The expected Paleogene poles of the reference point (30.0°N, 91.0°E) is 33.4° ± 2.9°N. (C) Equal-area projection of the late Eocene paleopoles of the central part of the QT. Small-circle fitting passes through 2 selected poles gives a paleolatitude of 11.2° ± 3.6°N (reference point: 30.0°N, 91.0°E). The expected Paleogene poles of the reference point (30.0°N, 91.0°E) is 33.4° ± 2.9°N. (D) Equal-area projection of Eocene paleopoles of the eastern part of the QT. The eastern part of the QT was situated at the paleolatitude of 23.9° ± 2.9°N in the period of 56.0–43.2 Ma. The expected Paleogene poles of the reference point (31.0°N, 98.2°E) is 35.5° ± 2.7°N.

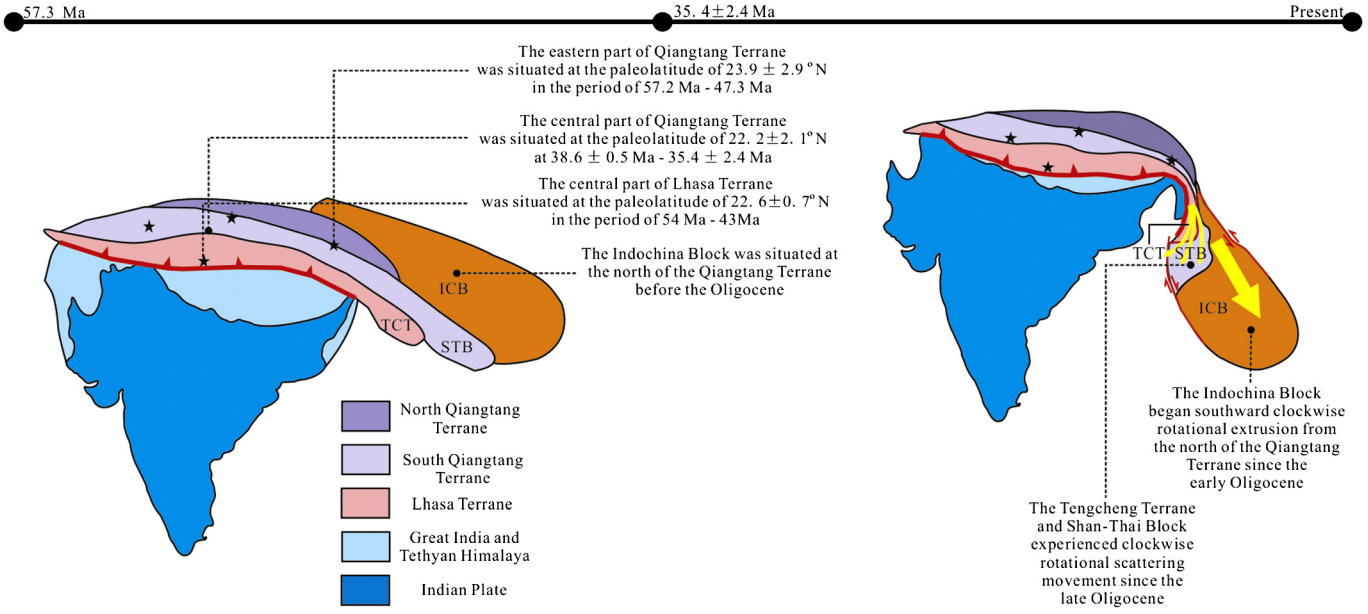


Fig. 7. The Cenozoic crustal movement characteristic of the southeastern part of the Tibet. The Indochina Block was situated at the north of the QT. Since the Oligocene – Early Miocene, the TCT and STB began southeastward clockwise rotational scattering movement in the form of ductile deformation. Since the early Oligocene, the ICB began to experience southward extrusion from the north of the QT, which absorbed part of the crustal shortening in the north of the QT.

sets (Lacassin et al., 1993; Leloup et al., 1995; Gilley et al., 2003; Wang et al., 2006; Clift et al., 2008; Lin et al., 2009). The Gaoligong strike-slip fault, as the western extrusion boundary, began right-lateral shearing contemporaneously, but it is difficult to estimate the total right-lateral offset (Wang et al., 2001; Searle et al., 2007; Lin et al., 2009). The consistent initial activation times of boundary faults indicate that the southward extrusion of the ICB and STB commenced no earlier than the Oligocene.

A paleomagnetic study of the Cretaceous red-beds in the northeastern edge of the LT suggests that a gap with a width of more than 1200 km remained between the LT and QT during the interval from the Cretaceous to the early Cenozoic (Otofuji et al., 2007). The STB and ICB were suggested to occupy this large gap, and subsequently were extruded southwards in the Cenozoic (Otofuji et al., 2007). However, the Paleogene paleomagnetic data for the QT and LT further indicate that they were situated at similar paleolatitudes at 35.4 ± 2.4 Ma, indicating that the gap had been closed before 35.4 ± 2.4 Ma. The LT and QT did not experience further crustal shortening after 35.4 ± 2.4 Ma, indicating that the southern part of Tibet could not provide sufficient crustal materials to accommodate the subsequent significant southward extrusion. The primary magnetic component obtained from the Eocene volcanic rocks in the TCT, and the early Miocene secondary magnetic components of the northern part of STB, show that the TCT and STB did not experience significant southward extrusion (Kornfeld et al., 2014; Tong et al., 2016). However, Cretaceous and Paleogene paleomagnetic studies and recent inclination tests of the ICB show that it experienced clockwise rotation accompanied by ∼700 ± 200 km southward extrusion since the Oligocene (Otofuji et al., 1998; Yoshioka et al., 2003; Tamai et al., 2004; Tanaka et al., 2008; Tong et al., 2013; An et al., 2017). Thus, it can be concluded that firstly the TCT and STB experienced intense crustal tectonic deformation and shortening in the Paleocene–Eocene, as also occurred in the LT and QT. Subsequently, since the Oligocene–Early Miocene, the TCT and STB began a clockwise rotational scattering movement, which induced right-lateral ductile shearing of the Gaoligong strike-slip fault. During this process, the crust of the TCT and STB moved southeastwards in the form of ductile deformation rather than significant southward extrusion, as was previously believed. Since 35.4 ± 2.4 Ma, the crustal shortening mainly occurred in the north of the QT (1300 km). It can be concluded that the ICB should have been situated in the north of the QT before the Oligocene (Fig. 7). Since the early Oligocene, the north of the QT began to experience enormous crustal shortening, and at the same time the ICB began to experience southward extrusion from the north of the QT, which absorbed part of the crustal shortening in the central and northern Tibet.

6. Conclusion

(1) A HTC was isolated from the Paleogene Gonjo Formation, in the eastern part of the QT. The tilt corrected site-mean direction of the Gonjo Formation is $D_s = 35.5°$, $I_s = 29.3°$, $k = 45.9$, $\alpha_{95} = 3.2°$, $N = 43$. Positive fold tests and a reversal test demonstrate that the HTC is of primary origin. Magnetostratigraphic analysis indicates that the HTC was acquired during the interval from 56.0–43.2 Ma. The inclination shallowing correction gave a paleopole of 57.9°N/192.1°E, $A_{95} = 2.9°$.

(2) The LT and QT experienced latitudinal crustal shortening before 54–43 Ma, indicating that southern and central Tibet experienced initial uplift in the Early Eocene. Subsequently, since 35.4 ± 2.4 Ma, ∼1300 ± 410 km of crustal shortening has occurred in the north of the QT, indicating that the northern Tibetan Plateau began to be uplifted since the Oligocene.

(3) Since the Oligocene–Early Miocene, the TCT and STB commenced a clockwise rotational scattering movement. In this process, the crust of the TCT and STB moved southeastwards in the form of ductile deformation rather than significant southward extrusion. The ICB was situated in the north of the QT prior to the Oligocene. Since the early Oligocene, the ICB began to experience southward extrusion from the north of the QT, which absorbed part of the crustal shortening in the central and northern Tibet.

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Appendix A. Supplementary material

Supplementary material related to this article can be found online at http://dx.doi.org/10.1016/j.epsl.2017.07.026.

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